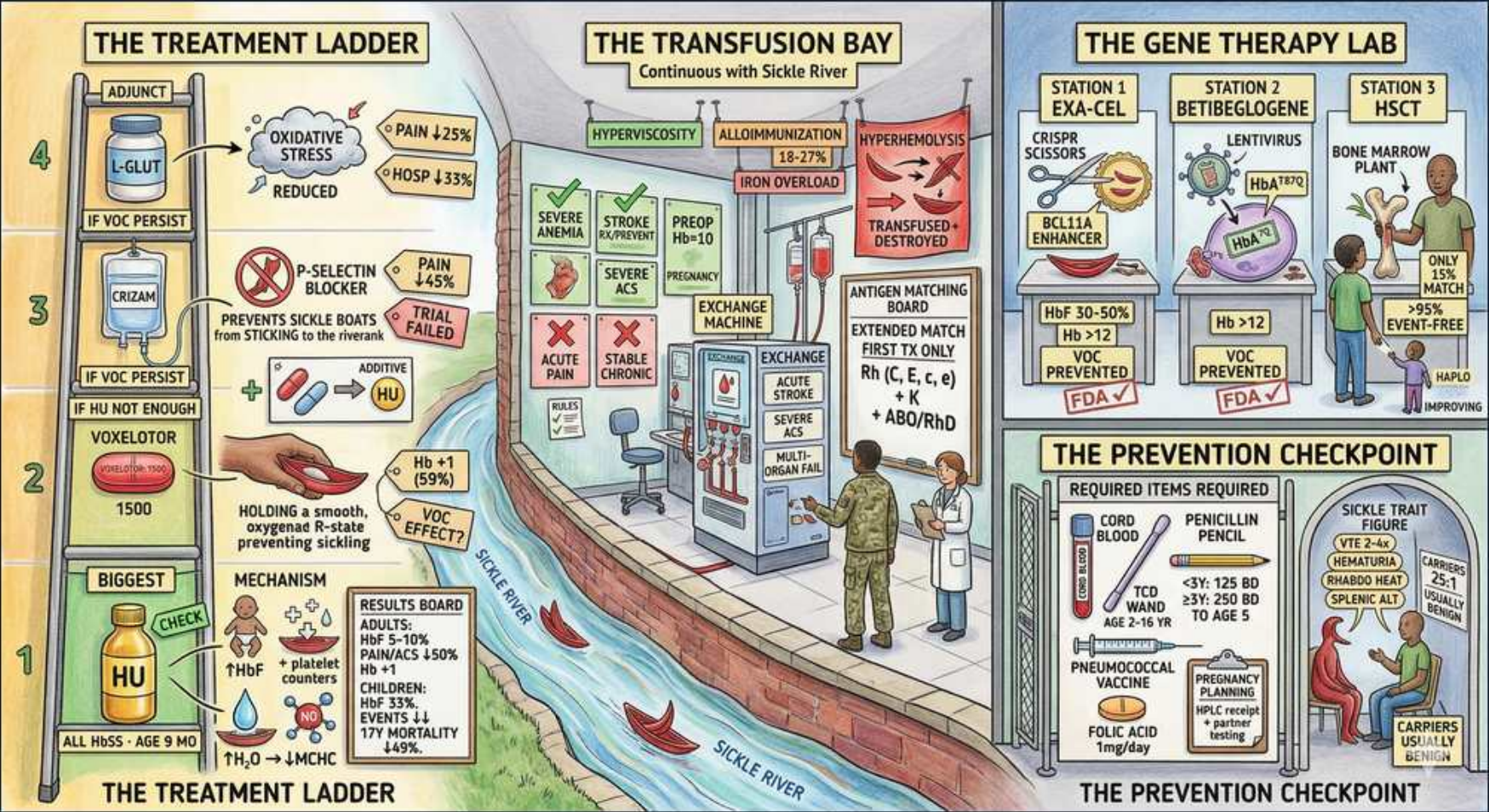


Hemoglobinopathy — The Sickie Armory



1. Hydroxyurea — base of ladder

WHAT YOU SEE

Step 1 'BIGGEST', HU bottle, ↑HbF, all HbSS age 9 mo+

WHAT IT ENCODES

Hydroxyurea is the first-line disease-modifying drug for SCD: it induces fetal hemoglobin (HbF), lowers WBC/platelet adhesion, and raises NO → fewer pain crises, less acute chest syndrome, fewer transfusions/hospitalizations, and reduced mortality (MSH and BABY HUG trials). Offer to ALL HbSS/Sβ0 patients starting at 9 months of age regardless of symptoms; titrate to max tolerated dose by ANC. Main toxicity: dose-dependent myelosuppression (cytopenias).

BOARD TRAP

Trap: reserving hydroxyurea for only severe/frequent-crisis patients. Discriminator: current guidance is to offer it to ALL HbSS infants from 9 months, not just symptomatic ones, because it reduces complications and mortality. Monitor the CBC for myelosuppression; it is teratogenic, so counsel on contraception.

2. Voxelotor

WHAT YOU SEE

Step 2 VOXELOTOR 1500

WHAT IT ENCODES

Voxelotor (1500 mg/day) is an HbS POLYMERIZATION inhibitor that binds the α-chain N-terminus and stabilizes the oxygenated, high-affinity R-state, keeping more HbS oxygenated so it cannot polymerize → raises hemoglobin level and lowers hemolytic markers. Used for chronic anemia. Note it primarily improves the Hgb number; its effect on reducing pain crises is limited.

BOARD TRAP

Trap: assuming voxelotor abolishes vaso-occlusive crises like an anti-adhesion agent. Discriminator: voxelotor's proven benefit is RAISING hemoglobin (anti-hemolytic, left-shifts O2 affinity), not robustly cutting pain crises — for crisis reduction think hydroxyurea or crizanlizumab. Because it increases O2 affinity, it can blunt the apparent benefit and complicate O2 unloading.

3. Crizanlizumab

WHAT YOU SEE

Step 3 CRIZAM, P-selectin blocker

WHAT IT ENCODES

Crizanlizumab is a monoclonal antibody against P-SELECTIN, an adhesion molecule on endothelium/platelets. By blocking P-selectin–mediated multicellular adhesion it interrupts the vaso-occlusion cascade → fewer/longer-interval vaso-occlusive crises (SUSTAIN trial). Given IV; an add-on for patients with recurrent crises despite/alongside hydroxyurea.

BOARD TRAP

Trap: confusing the mechanisms of the newer agents. Discriminator: crizanlizumab = anti-P-SELECTIN (blocks ADHESION → fewer pain crises); voxelotor = polymerization inhibitor (raises Hgb); L-glutamine = antioxidant. Matching drug to mechanism (adhesion vs polymerization vs oxidative stress) is the classic test point.

4. L-glutamine

WHAT YOU SEE

Step 4 adjunct L-GLUT, oxidative stress reduced

WHAT IT ENCODES

L-glutamine (oral) is an adjunct that boosts NAD redox potential in red cells, reducing OXIDATIVE STRESS and yielding a modest reduction in pain crises and ACS. It is added on, often with hydroxyurea. Its effect size is small relative to hydroxyurea.

BOARD TRAP

Trap: positioning L-glutamine as a first-line or stand-alone therapy. Discriminator: it is an ADJUNCT with modest benefit acting via reduced oxidative stress — never the answer over hydroxyurea for baseline disease modification, and not a crisis treatment.

5. Hydroxyurea mechanism

WHAT YOU SEE

Mechanism box: ↑HbF, ↓WBC, ↑NO, ↓45% mortality

WHAT IT ENCODES

Mechanistically hydroxyurea is a ribonucleotide reductase inhibitor whose dominant benefit is inducing HbF (which cannot enter the deoxy-HbS polymer); it also lowers leukocyte/platelet counts (less adhesion/inflammation) and increases nitric oxide. Net clinical effect in adults: roughly a 40% reduction in mortality plus fewer crises, ACS episodes, and transfusions.

BOARD TRAP

Trap: thinking hydroxyurea's benefit is mainly the drop in WBC/platelets. Discriminator: the PRINCIPAL mechanism is HbF induction; the cytoreduction and NO effects are secondary contributors. This is why it works across the whole sickle phenotype, not just the inflammatory component.

6. Transfusion indications

WHAT YOU SEE

Transfusion bay: severe anemia, severe ACS, preop, acute stroke

WHAT IT ENCODES

Transfusion (simple or exchange) is indicated for: symptomatic SEVERE anemia (incl. acute splenic sequestration, aplastic crisis), SEVERE acute chest syndrome, ACUTE stroke, acute multiorgan failure, and PREOPERATIVE optimization to Hgb ~10 g/dL for moderate/high-risk surgery. Chronic transfusion is used for primary/secondary STROKE prevention (keep HbS <30%). It is NOT used for routine uncomplicated pain crises.

BOARD TRAP

Trap: transfusing a patient in an uncomplicated VOC pain crisis. Discriminator: simple transfusion does not abort a pain crisis — reserve it for ACS, stroke, sequestration, severe symptomatic anemia, or preop. Over-transfusing toward Hgb >10–11 risks hyperviscosity, which can paradoxically worsen vaso-occlusion.

7. Transfusion hazards

WHAT YOU SEE

Hyperviscosity, alloimmunization 18-27%, hemolysis

WHAT IT ENCODES

Key transfusion hazards in SCD: HYPERVISCOSITY (keep post-transfusion Hgb $\leq 10\text{--}11$ g/dL), ALLOIMMUNIZATION (~18–27%; prevent with extended phenotype matching for Rh [C, E], Kell, and ABO), delayed hemolytic transfusion reactions / hyperhemolysis syndrome, and IRON OVERLOAD from chronic simple transfusion (treat with chelation — deferasirox/deferiprone/deferiprone).

BOARD TRAP

Trap: ordering standard ABO/Rh(D)-matched blood for a chronically transfused SCD patient. Discriminator: because alloimmunization is so common, use EXTENDED antigen-matched (Rh subtypes C/E, Kell) units. A patient who hemolyzes both transfused AND native cells after a transfusion = hyperhemolysis syndrome — give IVIG/steroids and AVOID further transfusion.

8. Exchange transfusion

WHAT YOU SEE

Exchange machine lowering HbS without iron load

WHAT IT ENCODES

Exchange transfusion (red-cell apheresis) rapidly lowers the percentage of HbS WITHOUT adding net iron, avoiding the iron overload of simple transfusion. It is preferred for ACUTE STROKE, severe/progressive acute chest syndrome, acute multiorgan failure, and as chronic therapy when iron overload must be avoided. Goal is typically HbS <30%.

BOARD TRAP

Trap: choosing SIMPLE transfusion for acute stroke or severe ACS. Discriminator: when you must quickly drop %HbS (stroke, severe ACS), EXCHANGE transfusion is preferred because it lowers HbS fast and limits iron loading. Simple transfusion is fine for sequestration/aplastic crisis/mild anemia where you just need to raise Hgb.

9. Gene therapy — exa-cel

WHAT YOU SEE

Station 1 EXA-CEL, CRISPR scissors, BCL11A enhancer, HbF >30-50%

WHAT IT ENCODES

Exa-cel (exagamglogene autotemcel / Casgevy) is a CRISPR-Cas9 ex-vivo gene-EDITING therapy that disrupts the BCL11A erythroid enhancer in the patient's own hematopoietic stem cells, de-repressing γ -globin to reactivate HbF (often >30–50%) → near-elimination of vaso-occlusive crises. FDA-approved (Dec 2023); requires myeloablative conditioning (busulfan) and autologous transplant.

BOARD TRAP

Trap: confusing the two gene therapies. Discriminator: exa-cel/Casgevy = CRISPR EDIT of the BCL11A enhancer to turn HbF back ON; lovo-cel = lentiviral ADDITION of an anti-sickling β -globin gene. Both use the patient's OWN cells (autologous) with myeloablative conditioning — so no GVHD, unlike allogeneic HSCT.

10. Gene therapy — beti-cel

WHAT YOU SEE

Station 2 lentivirus, HbAT87Q, Hb>12

WHAT IT ENCODES

Lovo-cel (lovotibeglogene autotemcel / Lyfgenia) uses a LENTIVIRAL vector to ADD a modified anti-sickling β -globin gene (β A-T87Q) into the patient's own stem cells, producing HbAT87Q that resists polymerization → higher total Hgb and fewer crises. FDA-approved (Dec 2023) for SCD; also requires busulfan conditioning and autologous transplant. (beti-cel/Zynteglo uses the same platform for β -thalassemia.)

BOARD TRAP

Trap: thinking lentiviral gene therapy carries graft-versus-host risk. Discriminator: because it is AUTOLOGOUS (patient's own edited cells), there is no GVHD and no donor needed — the trade-offs are myeloablative chemotherapy toxicity/infertility and a theoretical malignancy risk. Contrast with allogeneic HSCT, which DOES carry GVHD and needs an HLA-matched donor.

11. HSCT — curative

WHAT YOU SEE

Station 3 HSCT bone marrow, >95% event-free, matched sib

WHAT IT ENCODES

Allogeneic hematopoietic stem cell transplant from an HLA-MATCHED SIBLING is the only long-established CURE for SCD, with >90–95% event-free survival in children. Best outcomes when done early (before chronic organ damage). Limited by donor availability and risks of conditioning toxicity, infertility, and graft-versus-host disease/graft rejection.

BOARD TRAP

Trap: offering HSCT broadly as routine therapy. Discriminator: it is curative but limited by HLA-matched donor availability and carries GVHD/conditioning risk — historically reserved for severe disease with a matched sibling, ideally in childhood. The new autologous gene therapies are the alternative for patients lacking a matched donor.

12. Vaccines & penicillin

WHAT YOU SEE

Pneumococcal vaccine, penicillin pencil to age 5

WHAT IT ENCODES

Functional ASPLENIA (autosplenectomy from repeated infarction) leaves SCD patients vulnerable to ENCAPSULATED organisms (*S. pneumoniae*, *H. influenzae*, *N. meningitidis*). Give prophylactic oral PENICILLIN from age 2 months until at least age 5, plus the full vaccine set: pneumococcal (PCV + PPSV23), Hib, and meningococcal. Any FEVER in an SCD child is an emergency requiring immediate evaluation and empiric antibiotics.

BOARD TRAP

Trap: managing a febrile SCD child as an outpatient with watchful waiting. Discriminator: functional asplenia means overwhelming pneumococcal sepsis can kill within hours — fever mandates prompt cultures and EMPIRIC ceftriaxone, never delay. Skipping penicillin prophylaxis or vaccines against encapsulated organisms is the prevention error tested here.

13. Folic acid & screening

WHAT YOU SEE

Folic acid 1 mg/day; carrier counseling

WHAT IT ENCODES

Chronic hemolysis drives high RBC turnover, so give daily FOLIC ACID (~1 mg/day) to prevent megaloblastic/folate-deficiency crises. Comprehensive screening/prevention: universal NEWBORN screening (early dx), annual transcranial Doppler ages 2–16 (stroke prevention), annual dilated retinal exams, blood-pressure/renal (urine albumin) monitoring, echocardiography for pulmonary HTN, and genetic counseling for trait carriers.

BOARD TRAP

Trap: omitting folate supplementation or assuming a falling Hgb is always sickle-related. Discriminator: high cell turnover makes folate deficiency a real, preventable cause of worsening anemia — supplement routinely. Remember the surveillance bundle (newborn screen, TCD, retinal exam, albuminuria, echo) is part of standard preventive care, not optional.
